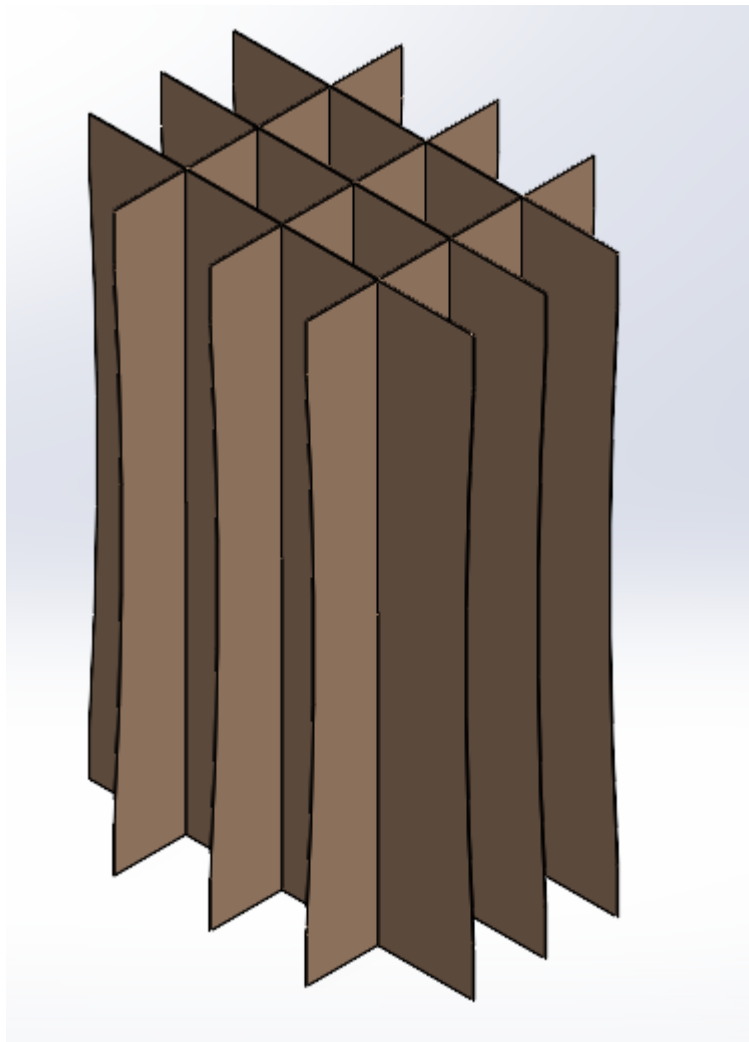


Sustainable Desk Challenge



MAE 191: Sustainable Engineering
Instructor: Huihui Qi, email: huqi@ucsd.edu

William Harris & Anya Ustinova
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1. Abstract:

This project aimed to design, test, and fabricate a scaled desk model that fits the required constraints. The challenge was to create a desk with a minimum height of 0.3 meters (12 inches) and a top surface of 0.2m x 0.15m (8in x 6in). The maximum weight was defined as 0.12kg (0.3 lb). The final design was required to support a minimum of 9kg (20lb) of loading. The material selection was also constrained to the provided [12x18-inch chipboard sheets](#). The team performed multiple FEA, topology, and trade studies and followed design for manufacturability practices to create multiple prototypes. The team designed, analyzed, and tested multiple prototypes in five weeks. The final design resembled an interlocked cardboard matrix. The final design passed the minimum weight requirement and structurally failed the final weight of 28 lbs. The performance index of the final design was calculated to be 98.25.

2. Introduction:

The Sustainable Desk Challenge required the team to design and fabricate a desk that could withstand the significant weight load while being as lightweight as possible. The desk was required to have a minimum height of 0.3 meters (12 inches) and a top surface of 0.2m x 0.15m (8in x 6in). The other requirement was to use only [12x18-inch chipboard sheets](#), with no glue or fasteners in the final design. The design is aimed to support at least 9kg (20lb). The project addressed the challenges and possibilities of creating sustainable design with limited materials and structural requirements. The project proved that even with limited resources and many constraints, creating a functional environmental design is possible.

This project addressed how the design of an everyday item like a desk could implement sustainable practices using limited materials. It focused on finding creative ways to reduce weight and create lightweight designs that could stay efficient without losing performance. By applying the weight-reducing strategies, the team developed a final lightweight design that was easy to assemble, test, and transport. The project highlighted the possibilities of innovating everyday household items like a desk to implement more sustainable practices.

The final report captures the methodology, design decisions, simulation results, testing, and prototyping of the Sustainable Desk Challenge. The report includes the results of the FEA analysis, topology studies, manufacturing rationale, and the final design justification. The paper also captures the prototyping methodology and discusses limitations and the assumptions made throughout the process.

3. Approach and Methodology

The official set of rules and constraints defined by the sustainable desk challenge can be seen in *Table 1*.

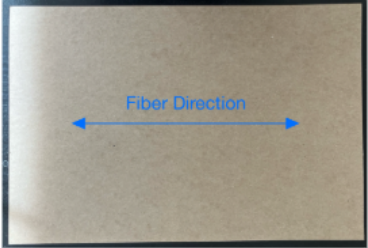
Elastic Modulus	3 GPa along the fiber direction, 0.5 GPa perpendicular to the fiber direction	
Tensile Strength	25 MPa along the fiber direction, 6 MPa perpendicular to the fiber direction	
Poisson's Ratio	0.3	
Density	682 kg/m^3	

Table 1: The Material Properties of the Provided sheet

Sustainable Desk Challenge Rules:

- Allowable materials: provided cardboard sheets only
- Minimum height: $H \geq 0.3m$ (12 in)
- Minimum length: $L \geq 0.2m$ (8 in)
- Minimum width: $W \geq 0.15m$ (6 in)
- Minimum load supported: 9kg (20lb)
- Maximum weight of your design: 0.12kg (0.3lb)
- Maximum deflection: $\Delta H \leq 10mm$ (0.4 in)

The team used the requirements and rules outlined above to transition into a rapid prototyping phase, where the focus was on iterating and refining the concept. From the beginning, the original idea was to create an interlocking structure, as it offered a practical balance of strength, simplicity, and material efficiency. Through three iterations, the team refined the design, ensuring it met both structural and functional needs while adhering to the project's constraints.

The Sustainable Desk Challenge followed a systematic approach to ensure the successful design and fabrication of a functional, sustainable desk. The methodology included the following key steps:

1. **Analysis:**

- Conducted a thorough evaluation of the project requirements, including dimensions, material constraints, and load-bearing capacity.
- Performed Finite Element Analysis (FEA) to simulate stress distribution and ensure structural integrity under the specified weight load.
- Conducted topology optimization studies to identify opportunities for material reduction while maintaining strength and stability.

2. **Design:**

- Developed initial design concepts based on structural efficiency and material constraints.
- Iteratively refined the designs through simulations and analyses, focusing on weight reduction and maximizing material utilization.
- Finalized the design to balance aesthetics, functionality, and sustainability.

3. **Prototyping:**

- Created a prototype using the specified chipboard sheets, ensuring precision in cutting and assembly.
- Tested the prototype for stability, load-bearing capacity, and ease of assembly.
- Documented observations to refine the design further, addressing any limitations or shortcomings.

4. **Manufacturing:**

- Assembled the desk without adhesives or fasteners, aligning with the sustainability goals.
- Tested the final product under real-world conditions to validate its performance.

This iterative process, combining design, analysis, and manufacturing, can be replicated using the tools outlined below.

The software and other tools that any team or an individual might need to attempt this project are:

- Design: CAD SOFTWARE (SolidWorks, NX, Fusion360, AutoCad)
- Analysis: Ansys, Solidworks
- Manufacturing: LaserCutter

4. Design & Results

The final design of the desk demonstrated good performance metrics, highlighting the efficiency of the interlocking structure and lightweight materials. Weighing only 0.12 kg, the desk was capable of supporting a load of 11.79 kg, (Fig 1) proving its structural integrity and ability to distribute weight effectively. Using the formula provided by the Sustainable Desk Challenge, the performance index was calculated to be 98.25 (*Weight Supported/ Weight of Structure*), reflecting the desk's excellent balance of strength and minimal material usage. This high-performance index underscores the success of the design in meeting the challenge's sustainability and efficiency criteria, making it a compelling example of lightweight, resource-conscious engineering. The results validate the design and prototyping process, demonstrating how simple, sustainable choices can achieve significant outcomes. All CAD and Simulation files can be found in the Appendix.

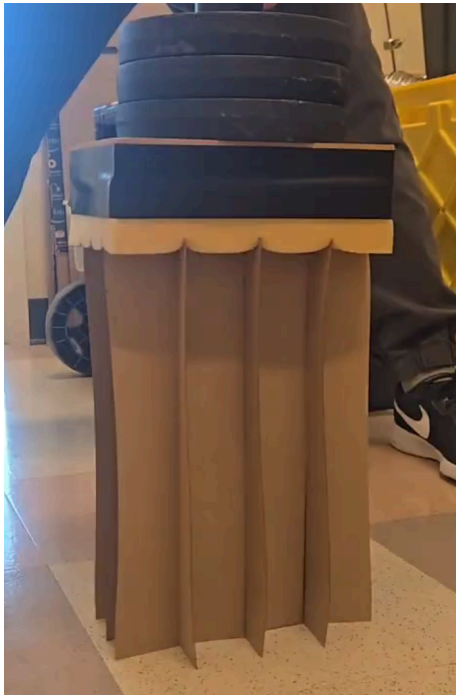


Fig 1. The desk under max loading

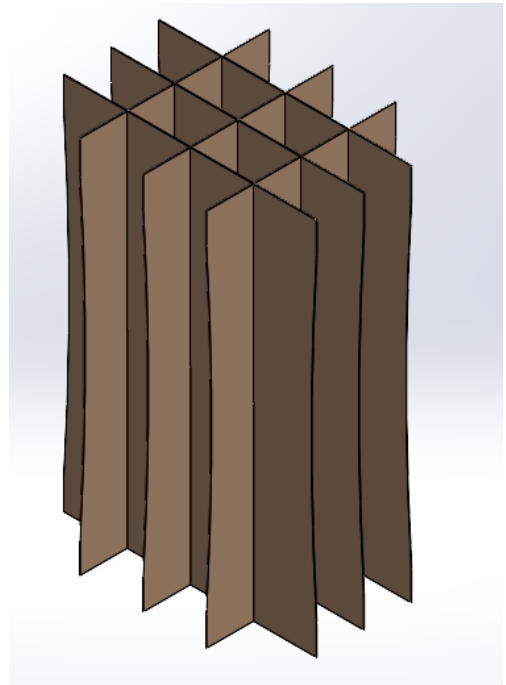


Fig. 2 Cad Model of the final design

The desk was made from interlocking sheets of cardboard, forming a grid-like pattern. (see *Fig.2*). The design consisted of vertical slits, allowing the sheets to slot together perpendicular to one another, creating a rigid and symmetric matrix. This modular structure provides uniform load distribution and stability through its vertical components. The interlocking design proved to be an excellent choice for the desk because it enhanced the structural integrity and evenly distributed the load across the entire framework while minimizing material usage. By slotting the cardboard sheets together perpendicular to one another, the design created a stable, self-supporting structure that was stable, even with lightweight materials.

From a manufacturing standpoint, the interlocking structure offered multiple advantages. The use of simple slots eliminated the need for adhesives or fasteners, reducing production complexity and waste. The modular design allowed for easy assembly and disassembly, making the desk both portable and recyclable. Furthermore, it can be flattened for transport if disassembly is not desired. The flat, 2D shapes could be efficiently fabricated using laser cutting, while the uniform components ensured scalability to different sizes, demonstrating the design's versatility and alignment with sustainability goals. (see *Fig 3*)

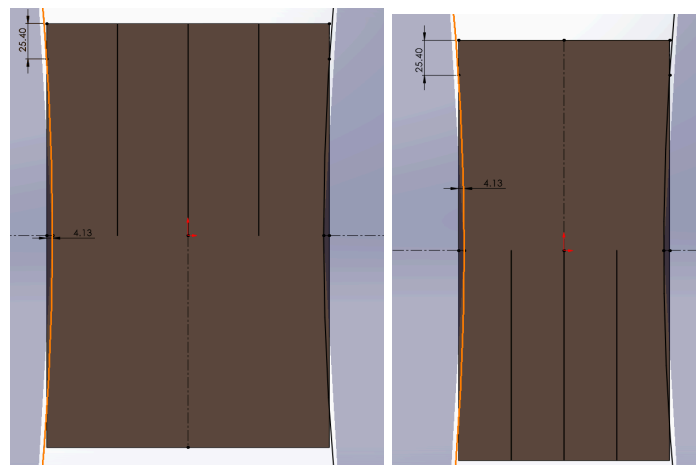


Fig 3. Two-sheet cardboard design

Moreover, to ensure the overall design was within the challenge's parameters, the team experimented with different methods of cutting weight from the design without compromising its structural integrity. Initially, this was done by filleting the corners of the sheets, which led to a weaker structure. This is because of the decreased surface area of the top and bottom faces of the design, which directly decreased the structure's load-bearing capacity. Therefore, the team opted to instead cut material from the outer edges of the sheets, which bear less load than the middle sheets. These cuts were made in a large arc along most of the sheet's height to

minimize the stress concentration, as seen in the simulation results below. (see *Table 2* and *Table 3*)

In this project, several assumptions were made to evaluate the performance of the desk design. The applied force was distributed evenly across all faces of the desk's top surface to simulate a uniform load, representing a realistic use case for the desk. Fixed supports were applied only to the middle section faces, corresponding to the load-bearing members of the interlocking structure. This setup allowed for an accurate analysis of how the desk would handle weight under practical conditions while maintaining boundary conditions that reflected its actual use. (See *Fig.4*)

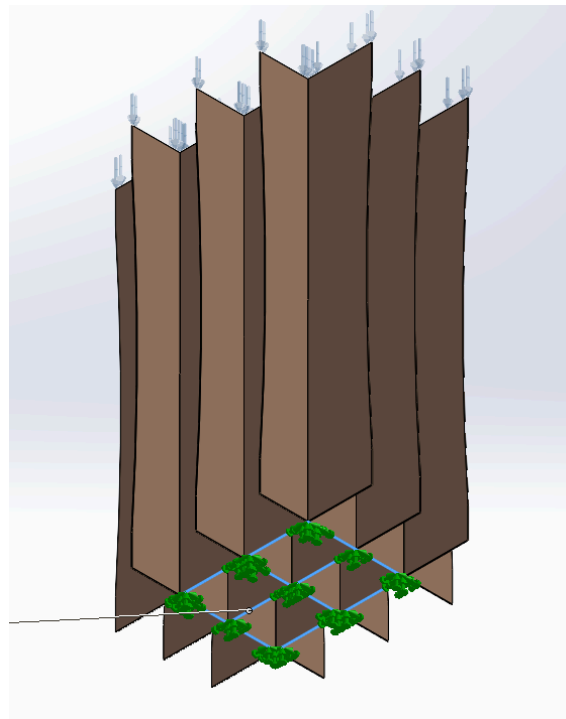


Fig 4. Fixed supports and load application during analysis. Green arrows indicate fixed support locations at the base, while grey arrows show the distributed load application.

Material properties were assumed to be isotropic due to the lack of sufficient data to model anisotropic behavior. While this assumption simplified the simulation, it did not fully capture the complexities of the cardboard material, which had anisotropic characteristics due to the direction of the fiber. Screenshots of the assumed material properties were stored in the shared drive for reference and can be seen in the Appendix.

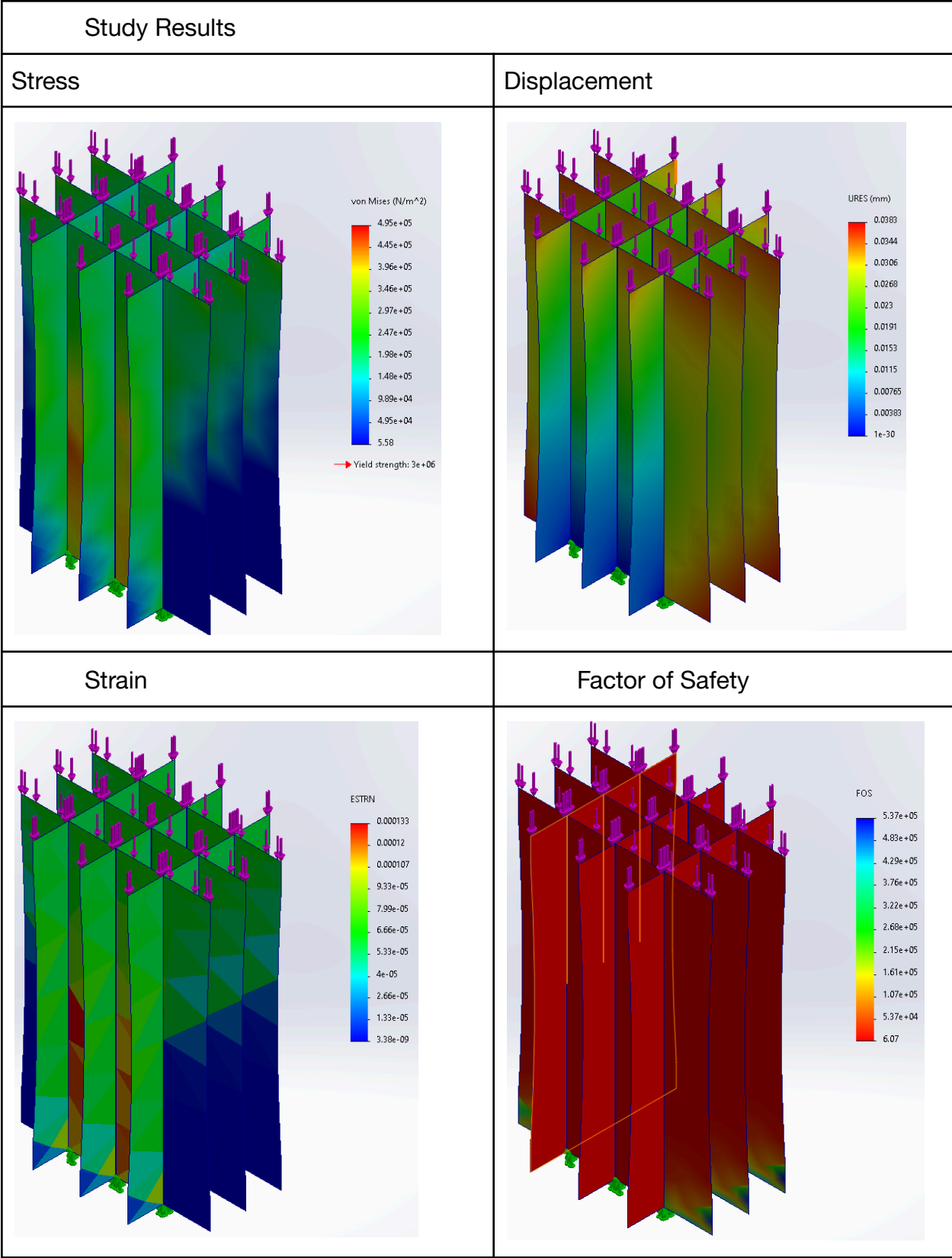


Table 2: Simulation Results supporting 27.54 lbs (12.49 kg)

As can be seen in Table 2, The FEA study results validate the effectiveness of the interlocking matrix structure's load distribution. The von Mises stress analysis shows peak stresses of approximately $4.95 \times 10^5 \text{ N/m}^2$ concentrated in the lower half of the middle sheets, close to the interlocking joints, well below the material's yield strength of $3 \times 10^6 \text{ N/m}^2$. The strain distribution follows similar patterns with peak values of 0.00013 near the joints. Maximum displacement occurs at the top surface edges of approximately 0.04mm, with a minimal deflection at the base, which aligns similarly to what we see practically; however, these displacements are higher in reality.

Most significantly, the factor of safety maintains a minimum value of 6.07 throughout the structure, validating that the design can safely handle loads beyond the minimum requirements. The uniform distribution of safety factors across components confirms the effectiveness of the interlocking matrix approach for load distribution. However, while the factor of safety remains relatively high when the structure supports 27.54 lbs (12.49 kg), the structure seems to fail when it hits 27.65 lbs (12.54 kg). This is probably due to the displacement at the edges, causing the structure to bend and quickly experience higher stress concentrations. This is consistent with what was observed in physical testing, pictured below in Table 3 and captured on video, linked in the appendix.

The desk was assessed using a performance index formula:

$$\text{Performance Index} = \frac{\text{Max Load Supported}}{\text{Weight of your Design}}$$

The theoretical performance index, according to the simulation was 104.08, while the practical performance index was 98.25 (94.4% of the theoretical index). This demonstrates a close alignment between simulation predictions and real-world performance. However, it is noteworthy that the design withstood 26 lbs and failed at 28 lbs in real life, which means it experienced a higher load than it was theoretically capable of. Additional tests with weights of 27 lbs and 27.5 lbs would provide a more accurate account of the desk's load-bearing capabilities and performance index. These tests are ideal. However, under time constraints, they may not be possible.

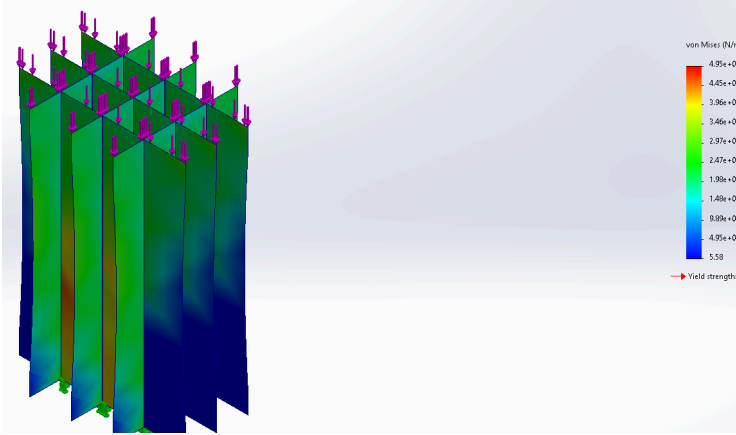
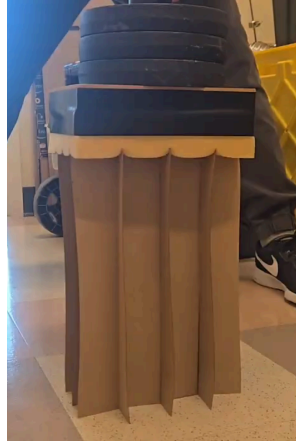
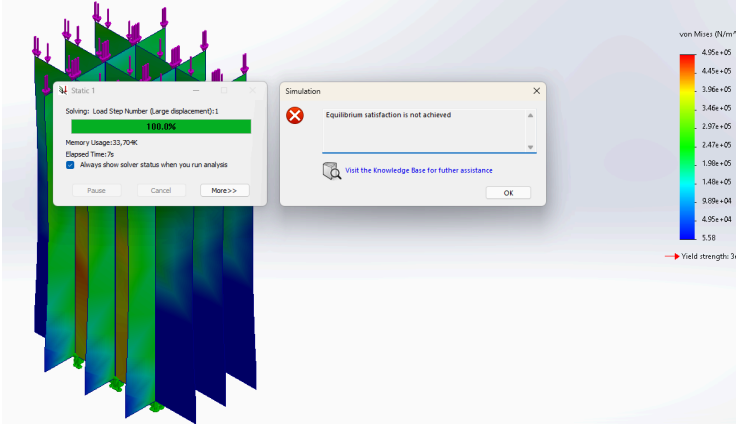
	Simulation	Reality
Success	 122.5 N (27.54 lbs) (12.49 kg)	 26 lbs (11.79 kg)
Failure	 123N (27.65 lbs) (12.54 kg)	 28 lbs (12.7 kg)
Performance index	$PI = \frac{12.49}{.12} = 104.08$	$PI = \frac{11.79}{.12} = 98.25$

Table 3: Failure Threshold in Reality vs Simulation

Prototyping Methodology

The initial design for the desk drew inspiration from existing cardboard furniture and origami structures widely used worldwide. A notable source of inspiration was the "[Seat 8](#)," a cardboard chair design created by students at the University of British Columbia. This design featured a simple structure of eight interlocking cardboard sheets arranged in a matrix pattern. Adopting a similar approach, the team developed an initial concept with a desktop of a folded sheet secured with interlocking tabs and

slots. However, this design exceeded the maximum weight requirements set by the project parameters. (see *Fig 5*)

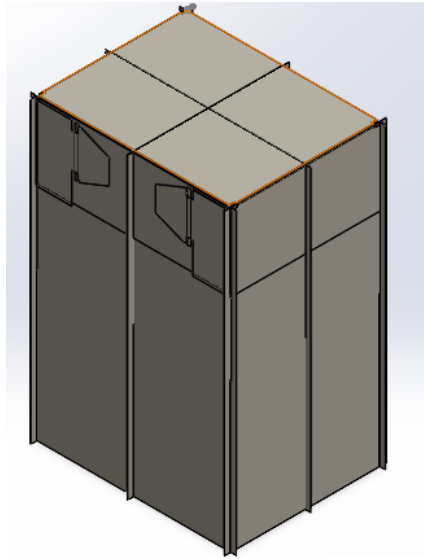


Fig.5 The first prototype

Additionally, the team identified potential failure points in the origami-style folds of the desktop, which could compromise the structure under load. Testing videos and images further highlighted the weaknesses of this approach, prompting a redesign to meet performance expectations.

In the second design phase, the team aimed to ground their approach in empirical data by leveraging the material properties of clipboard sheets to perform finite element analysis (FEA). Due to limited information, several assumptions about the material's behavior had to be made, reducing the simulations' accuracy.

The team began by running a topology study on a solid block measuring 0.15 x 0.2 x 0.3 m, with the bottom face fixed and a load applied to the top. While this initial model treated the material as a solid block—an unrealistic assumption given the sheet-based nature of the actual material—it provided insights into efficient material distribution. The output suggested removing most of the block's mass, leaving only a solid top surface and two diagonal "legs" connecting the bottom corners to the top center. (See *Fig. 6*) Though clipboard sheets could not directly replicate this structure, the results inspired the idea of using five legs, one at each corner and one in the center, to balance load distribution and meet the weight requirements.

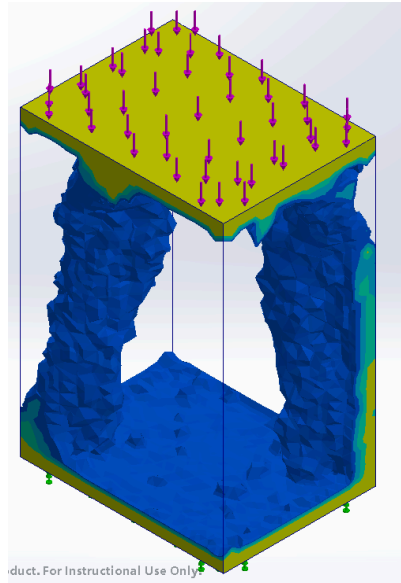


Fig. 6 Topology optimization results of a solid block based on 20lb loading

Building on these insights, the team conducted another topology study, this time using a matrix structure to better approximate the material's sheet-based behavior. By applying a load to the top face and fixing the middle and four corners of the bottom face, the simulation produced results that informed the next phase of design. (see Fig. 7)

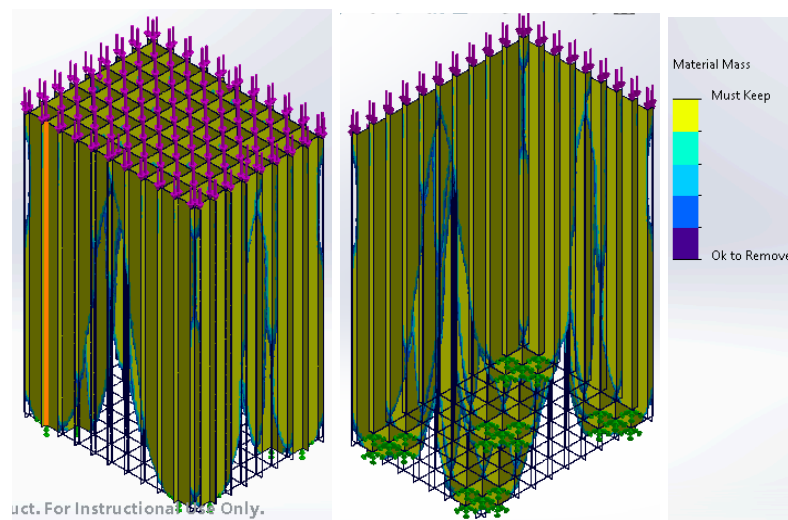


Fig. 7 Topology results of matrix structure with five fixed supports

The team explored various methods for constructing the five legs within the project's constraints. The primary idea was to fold the cardboard sheets into load-bearing columns with different shape profiles, such as square, triangular, or circular prisms. Several concepts were developed, including folding sheets into

triangular or square prisms and interlocking them with additional components to compensate for the thin columns' lack of stability. While a five-legged design could theoretically support a higher load, the team concluded that such a design was impractical under the challenge's parameters because more material is needed to prevent the desk from failure due to torsion. (See Fig. 8)

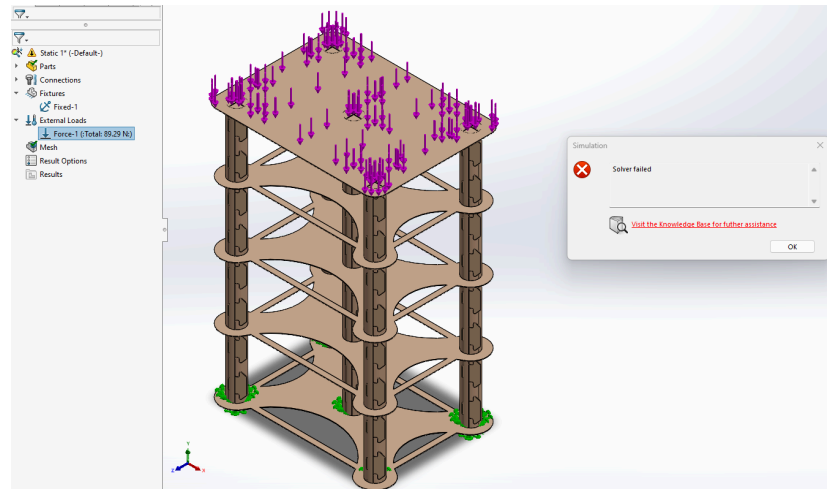


Fig. 8 The five-legged design

After looking into other designs, the team revisited the initial grid-like concept. The team carefully considered how to improve the design best to remove the flaws that initially caused concern. Previously, the slits in the sheets, which allowed them to interlock, were made to be close to the edges of the sheets, which would cause tearing and the desk to collapse under minimal loading. Furthermore, the design incorporated a folded sheet for the desktop, interlocked with the vertical sheets. The team was concerned about potential instability caused by the folds in the material if such a part were to be used for structural support. These concerns led to the final design, which removed the desktop altogether and opted to symmetrically apply the slits in the sheets to be at each fourth of their length to ensure stability.(See Fig 1, Fig2, and Fig 3)

6. Discussion

The findings highlight the effectiveness of the interlocking design for lightweight, sustainable structures. The desk, weighing just 0.12 kg, demonstrated an impressive ability to support a load of 11.79 kg, achieving a performance index of 98.25 as calculated using the formula from the Sustainable Desk Challenge. These results validate the potential of this design approach to create strong yet resource-efficient structures, showcasing how limited materials can be used effectively to meet both

functional and sustainability goals. The study also demonstrates how thoughtful design and simulation practices can optimize performance while minimizing material waste.

This project addressed the challenge of using limited materials to create a lightweight yet load-bearing desk. By employing an interlocking design and iterating through multiple prototypes, we achieved a balance between strength, material efficiency, and ease of assembly. Additionally, the project solved the issue of replicability by providing a straightforward design and prototyping process supported by accessible tools and software, which others can adopt to create similar structures.

The methodology and findings from this project can serve as a framework for others in sustainable design and engineering. Using CAD software for design, simulation tools for analysis, and laser cutters for manufacturing creates a streamlined process that is easy to replicate. Additionally, the demonstrated success of the interlocking structure can inspire further applications in areas such as packaging, temporary furniture, and structural frameworks. The detailed documentation of material properties, assumptions, and testing conditions also provides a valuable reference for anyone exploring similar lightweight, sustainable solutions.

While the study provided valuable insights, there were several limitations. The material properties were assumed to be isotropic due to a lack of data, which may have resulted in slight inaccuracies, as cardboard often exhibits anisotropic behavior. Additionally, the simulations and testing were limited to a load of 11.79 kg, and while the table performed well under this weight, further testing with heavier loads (e.g., 27 lbs and 27.5 lbs) would provide a clearer picture of its maximum capacity. Finally, the environmental durability of the structure, such as its performance under moisture or repeated loading, needed to be evaluated, leaving room for future studies to address these aspects.

6. Conclusion

This project demonstrated how thoughtful engineering can deliver sustainable solutions from simple designs and basic materials. The final desk design successfully met its primary objectives: supporting substantial loads, while maintaining minimal weight and relying solely on clipboard sheets. The interlocking matrix structure proved particularly effective, offering strength and stability, without the need for adhesives or fasteners.

In addition to the success of the final design, the project's approach demonstrates significant potential for scaling to other applications. The documented methodology, which combined CAD modeling, empirical testing, and iterative

prototyping, serves as a framework for others to replicate and adapt. This streamlined process allowed for rapid development and optimized material usage, aligning with the project's sustainability goals.

Looking ahead, future work should explore areas for improvement, such as incorporating more precise material data to refine simulations and better capture the properties of the cardboard. Moreover, testing under diverse conditions, such as exposure to moisture or repeated loading, could provide valuable insights into the design's durability and practical performance. By addressing these aspects, the interlocking matrix concept could evolve into a versatile solution for a broader range of structural and sustainable applications.

7. Appendix

1. Material Properties

Isotropic (Used)	Anisotropic (Unused)																																																																																																															
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Compressive Strength in X		N/m ²																																																																																																														
Compressive Strength in Y		N/m ²																																																																																																														
Shear Strength in XY		N/m ²																																																																																																														
Yield Strength	6000000	N/m ²																																																																																																														
Thermal Expansion Coefficient in X		/K																																																																																																														
Thermal Expansion Coefficient in Y		/K																																																																																																														
Thermal Expansion Coefficient in Z		/K																																																																																																														
Thermal Conductivity in X		W/(m·K)																																																																																																														
Thermal Conductivity in Y		W/(m·K)																																																																																																														
Thermal Conductivity in Z		W/(m·K)																																																																																																														
Specific Heat		J/(kg·K)																																																																																																														
Material Damping Ratio		N/A																																																																																																														

- [Desk Failure at 28 lbs](#)
- [Sustainable Design Challenge: Desk](#)
- [CAD Files](#)

Category	Description	Point Breakdown
Cover Page & Table of Contents	Professional cover page which includes the following information: - Project title - MAE 191: Sustainable Engineering - Instructor: Huihui Qi, email: huqi@ucsd.edu - Team member names - Date Include a table of contents	3%
Abstract	Concise technical summary of your project - this should provide readers all the top-level information they expect to find in your paper, including what you've done and what results you have obtained.	10%

Category	Description	Point Breakdown
Introduction	Rationale and motivation: - What is the reason for carrying out this project? - What problem does the project aim to solve?	5 %
	Project goal and scope: What does your overall project aim to address?	5 %
	Overview of project: - For analysis, design, and manufacturing (if applicable) - Outline structure of your paper	5 %
Main Body: Project Study & Results	Approach and Methodology - Summary of how you conducted your project or LCA, sufficient for others to replicate - Include software, tools, and other relevant information	10 %
	If your project is LCA: - Goal & scope of LCA study - Simplifications and assumptions - Findings and results of your study, including impacts and interpretation of results - Comparison involving your design/product - Data should be clearly presented - Critical figures/illustrations should be presented in the main body	25 %
	If your project involves designing of fabricating a product (including the cardboard desk challenge): - Demonstrate function (physically or digitally) - Justify “design” decisions; this could be for the design of the product and/or associated processes - Evaluate the outcome based on expectations; explain any deviations or discrepancies - Critical figures/illustrations should be presented in the main body	
Detailed Discussion and Conclusion	Meaningful, in-depth discussion of your project results/outcomes, including merits, impacts, and limitations: - Significance of your findings - What problem(s) were you able to solve? - How can your work be used by others in the field? - Address shortcomings of your study - Consider your approach and methodology - Must reference assumptions and simplifications - Uncertainties in results	20 %
	Room for future work: - Unanswered and newly-raised questions - How can your work be used by others in the field?	5 %
	Conclusion	5 %
Organization, Polish, and Format	Clarity of writing, including visuals/figures, data, organization, logical progression, appendix, etc.	7 %

